

Physics-informed framework approach in predicting the dynamics of Ising-like Zimm-Bragg systems

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(Dated: February 16, 2026)

This project focuses on developing a Physics-Informed Neural Network (PINN) framework to model the conformational dynamics of Zimm-Bragg systems, which are essential for understanding helix-coil transitions in biophysics. The study aims to integrate the statistical mechanical constraints of the Zimm-Bragg model into a deep learning architecture to improve prediction accuracy while reducing the need for massive experimental datasets. This approach ensures the tool doesn't just recognize patterns in data but actually sticks to the underlying biophysical principles of the Zimm-Bragg model, leading to more reliable and scientifically grounded results.

I. INTRODUCTION

The Zimm-Bragg model serves as a fundamental theoretical model for describing how polypeptides transition between helical and random-coil states based on cooperative interactions. While traditional computational treatments like Monte Carlo simulations are accurate, they are often computationally intensive and struggle to scale with complex biological environments.

Current machine learning approaches often treat these transitions as "black box" problems, which can lead to results that violate known physical principles or fail when data is sparse. This project addresses that gap by creating a hybrid framework that enforces physical consistency, making the model both efficient and scientifically grounded. To achieve this, we use a hybrid approach that integrates the Zimm-Bragg model directly into the learning process. By using closed-form equations achieved via the hydrodynamic approximation we can constrain the model's behavior and gain immediate insight into how cooperativity and nucleation drive the system. This analytical method allows for rapid testing across various conditions without the high cost of full-scale simulations. To ensure accuracy, we use detailed simulations as a "golden standard" for validation, capturing the physical nuances overlooked in the hydrodynamic approximation. This framework ensures high efficiency without sacrificing predictive reliability.

II. RESEARCH QUESTIONS

The primary goal is to test the applicability of the hydrodynamic approximation as a mediator for structural dynamics prediction. We must find specific architectural constraints that effectively map the discrete states of a Zimm-Bragg system onto a continuous neural network optimization space such that the thermodynamical properties of the system are conserved.

Another key question is whether including physical constraints allows the model to stay accurate when testing on temperature ranges or sequence lengths it hasn't seen during training.

III. METHODOLOGY

The methodology begins with a literature review to study the biophysical basis of the model and simulations and identify the shortcomings of existing predictive techniques. I will then generate highly accurate synthetic data using established Monte Carlo simulations to serve as the ground truth for training.

The core development involves building a custom neural network loss function that penalizes predictions deviating from the physical free energy dynamics provided via the hydrodynamic approximation

$$\begin{aligned}\dot{m} &= -(1 - a_1)m + \frac{a_0}{2} + \frac{a_2}{2}\epsilon_2, \\ \dot{\epsilon} &= -2\epsilon + a_1 + (a_0 + a_2)m + a_1\epsilon_2,\end{aligned}\tag{1}$$

where a_0, a_1, a_2 are composite constants of the thermodynamics, m is the helicity degree, ϵ is the correlation in of the Zimm-Bragg chain and ϵ_2 is the second neighbor correlation found as a function of m and ϵ via the hydrodynamic approximation.

Finally, the efficiency of this "physics-informed" approach will be evaluated by comparing its accuracy and computational cost against traditional sequence models like KOTs (Koopman operator theory based mathematical frameworks).

IV. EXPECTED RESULTS

The project will deliver a validated physics-informed neural network framework capable of predicting Zimm-Bragg dynamics with physical consistency. By incorporating thermodynamic laws directly into the model's training, the framework will follow the actual biophysical

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rules of the Zimm-Bragg model rather than just mimicking patterns in the data. I also expect to produce a set of guidelines for researchers on how to embed similar Ising-like constraints into other data science problems involving sequential biophysical data.

V. TIMELINE AND DELIVERABLES

The proposed research will be conducted over a twelve-week period, structured into distinct phases to ensure a smooth progression from theoretical formulation to final validation.

Task Description	Timeline	Deliverable
Phase 1: Literature & Theory		
1.1 Literature review of Zimm-Bragg	Weeks 1–2	—
1.2 Loss function formulation	Weeks 1–2	Theoretical model
Phase 2: Data Acquisition		
2.1 Synthetic data generation	Week 3	Pre-processed dataset
Phase 3: PINN Development		
3.1 Model architecture design	Weeks 4–5	Training scripts
3.2 Loss function integration	Week 6	PINN prototype
Phase 4: Comparative Analysis		
4.1 Baseline implementation	Weeks 7–8	Benchmark results
4.2 Performance evaluation	Weeks 9–10	Comparative metrics
Phase 5: Finalization		
5.1 Visualization and error analysis	Week 11	Figures and plots
5.2 Final paper submission	Week 12	Research paper

TABLE I. Timeline and deliverables

VI. RISKS

A significant risk is the potential for the physics-based loss term to create “vanishing gradients,” making the model difficult to train. I intend to minimize this by using adaptive weighting for the loss components and starting with simplified one dimensional systems. Another risk involves data collection issues if simulations do not capture edge case transitions, which if needed, will be addressed via importance sampling. Finally, there is a risk of project delays due to the high computational cost of physics-based training. To manage this time constraint, tasks are prioritized to ensure core functionality: the primary focus remains on the model and basic adaptive weighting, while high-dimensional scaling and exhaustive hyperparameter optimization will be less focused on if time is running short.

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